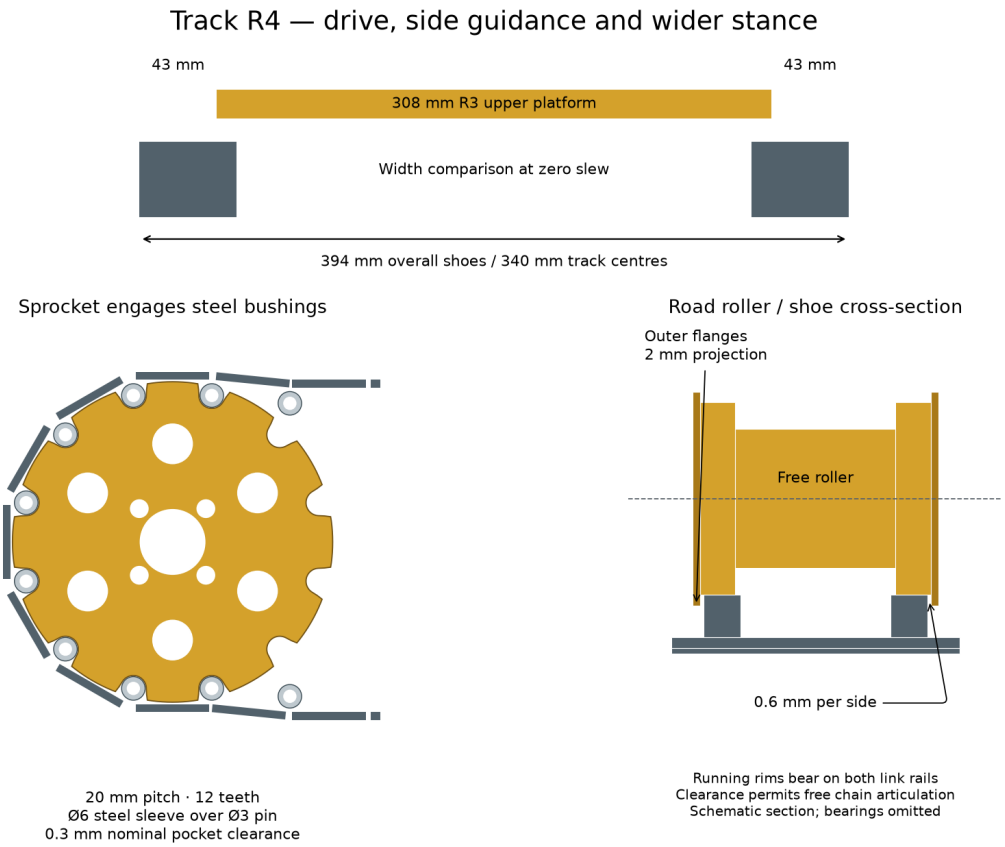
Tracked arm60 — build guide



Track R4 interfaces



This is a digital prototype for a later physical build. The supplied arm is the unchanged R4 arm60, with four standard V6-U60 actuators. Printer size and splitting are deliberately deferred. Do not scale the meshes to fit a printer: doing so would change bearing seats, gear pitch and the existing arm interfaces.

Materials and print preparation

Use PETG for the chassis, decks, cab and links. Use a tough, accurately printed PETG or PA gear set; the calculations use an assumed 8 MPa allowable tooth-root stress, which must be confirmed for the actual material and print process. No steel shaft, track pin, bearing, clamp hub, spacer or ballast is a printed substitute.

The STL files are in millimetres and placed on Z=0. Rails, bearing covers and wheels are laid with their Y axes vertical. The deck, counterweight, engine hood, tank housing, toolbox, equipment tray and motor saddles are inverted to put their broad flat faces on the bed. Support the stepped regions under the hood nose and the toolbox step in the slicer. Track R4 shoes have three 1 mm grousers and print on their tread faces; check support and bridging across the recessed grooves in the slicer. Print one ordinary and one `_left` rail/carriage; the supplied left meshes are already mirrored. All other quantities are in the BOM.

Start with `22_fit_coupon.stl` and two track shoes. Establish the actual hole compensation before printing the vehicle. The coupon also carries a Ø19.7 mm 18650 groove: a real cell must drop in and turn freely. Suggested starting settings are a 0.4 mm nozzle, 0.20 mm layers, six perimeters and dense local regions around holes and gear teeth. The mass calculations use the complete modeled PETG volume, not a particular slicer infill percentage. Reweigh the actual assembly and repeat the stability calculation if the mass distribution changes substantially.

The nominal 608/6004 seats are 22.2/42.2 mm; they are clearance seats retained by shoulders and fasteners, not a promised press fit on every printer. Tune them using the coupon. Ream running pin holes to 3.2 mm and axle clearances to the drawing. Deburr every track knuckle. Check flatness of the bearing-housing flange and decks. The original monolithic boom is approximately 414 mm long; splitting it is a separate engineering task.

Track drive and shafts

The two motors have independent H-bridge channels. Both tracks moving forward gives straight travel; opposite directions gives a skid turn. Left and right motor polarities must be calibrated after assembly because their shafts face opposite ways.

Each sprocket has 12 rounded bushing pockets at 20 mm pitch. Its pitch radius is 38.637 mm and tooth-tip radius is 40.821 mm. Fit one Ø6 OD × Ø3.1 ID × 17 mm steel sleeve over each Ø3 pin between the inner link ears. The sprocket drives the sleeve; the plastic knuckles stay outside its faces. Deburr both sleeve ends and check the free slip fit before assembly. This captured, replaceable sleeve adapts real pin-and-bushing chain construction to the model; it is not a full-size sealed track joint.

The root pocket has 0.3 mm nominal radial clearance and swept entry/exit relief calculated for rigid 20 mm links. Tooth tips clear the shoe web by 0.5 mm nominally. Six open web windows distinguish the toothed rim and bolted hub. Do not replace this profile with closed round holes or shallow circular scallops. See the Berco reference photos and explanation in `references/TRACK_SYSTEM.md`.

For each drive shaft, working from the machine centre outward (right-hand coordinates shown; mirror Y for the left):

Component	Y interval, mm	Notes
37 mm motor body	19 to 78	Maximum body envelope; shaft Ø6 D, 78 to 92
Flexible Ø6-to-Ø8 coupling	82 to 102	Ten millimetres of engagement in each bore; both bores clamp
Ground Ø8 steel drive shaft	92 to 184	92 mm long; no wheel load through the motor bearings
Ø8 shaft collar	104 to 109	Retains inner bearing; coupling has 2 mm clearance to collar
Inner 608 bearing	109 to 116	Housing cover at 106 to 109
Steel inner-race spacer	116 to 133	OD12, ID8.1, 17 mm long
Outer 608 bearing	133 to 140	Housing cover at 140 to 143
Steel spacer	140 to 149	OD12, ID8.1, 9 mm long
Purchased clamp hub	149 to 163	Ø8 shaft bore; Ø34 flange at 160 to 163
Printed sprocket	163 to 177	Four M4 clearance holes on Ø24 pitch circle

The motor saddle carries the upper half of the motor body. A formed steel strap closes its lower half and bolts to the bridge with two M4 through fasteners. Cut/form the strap to the STEP envelope, deburr it, and use a thin insulating/friction liner if needed. Do not rely on an unsupported printed D-hole or a set screw threaded directly into plastic to transmit drive torque.

Each free wheel has two 608 bearings, a steel inner-race spacer and shaft collars. The road/return rollers use a 26 mm inner spacer, 10 mm inboard spacer and 1 mm outboard washer. The idler uses a 6 mm inner spacer, 20 mm inboard spacer and 11 mm outboard spacer. All have OD12/ID8.1. These spacers contact inner races only; the 12.4 mm body bore gives running clearance. Tightening collars must not make a wheel stiff. Free shafts are Ø8 × 79 mm (Y=117 to 196); collars are Ø8 bore × 5 mm long.

Road rollers are at X=-52, 0 and 52, Z=29. Return rollers are at X=±35, Z=63.641. The nominal drive/idler centres are X=-110/+110, Z=46.321. The sprocket and idler are raised 2 mm above the central ground-run geometry. This keeps the lower rollers as the main chassis supports. To preserve exact pitch, the assembly idler sits 0.201 mm behind its nominal X110 position (X109.799); top and bottom transition links slope between the end-wheel chords and central runs. Tracks are centred at Y=±170. Steel track pins are Ø3 × 56 mm with grooves for the chosen retaining clips at Y=±26.5 relative to each track centre. Use 68 pins, 68 steel sleeves and 136 clips; groove width/depth must follow the actual clip supplier. The clip envelope is shown, but individual groove/clip details are not manufacturing geometry. Deburr the pin ends and keep them free of threads over the bearing surfaces.

Idler adjustment and polygonal motion

Use the R4 idler with twin rail rims and outer guide flanges on every idler, road roller and carrier roller. The running surfaces support the two link rails. Guide faces sit at ±21.6 mm from the track centre around the ±21 mm rail faces: 0.6 mm clearance on each side, with 2 mm flange projection. The wheels guide with clearance; they do not squeeze the chain. Compliance comes from the hinges and sprung idler take-up. Check both side clearances and clip-to-flange clearance while moving the chain sideways through its available play and hand-driving a complete circuit.

The wider shoes are 54 mm. Overall shoe width is 394 mm versus the 308 mm R3 upper platform, 43 mm beyond each side; the pin envelope is 396 mm. Frames move outward by 16 mm and wheel centres by 20 mm per side, so the R4 bridge, shafts and spacers must be used together. Drive-wheel overhang from the outer bearing centre is now 33.5 mm. The supplied calculation uses this increased bending arm.

Join 34 shoes per side. The complementary hinge ears must alternate as shown in the STEP assembly. Fit the last pin with the idler retracted. A 12-tooth sprocket has approximately 3.4% chordal speed variation; the chain must not be tensioned like an inextensible smooth belt.

The idler carriage has -3 to +6 mm of assembly adjustment. Its M5 stud and captive nut pull it forward through a compression spring outside the front reaction plate. The nominal spring envelope is OD9, ID at least 5.4, 12 mm length at nominal X110, or 11.799 mm in the exact-pitch assembly pose. Select a measured spring near 4 N/mm with 15 N force at nominal X110 and about 15.8 N in the assembly pose (approximately 15.75 mm free length), sufficient usable stroke and no coil bind over the permitted travel. Its CAD tube is an envelope, not a printable solid spring.

Two M4 shoulder fasteners guide each carriage through the rail slots. Set their shoulder/spacer stack for 0.3 mm sliding clearance; tightening ordinary screws hard against the carriage would defeat the spring. Use washers and locking nuts, then confirm smooth travel with no lateral wobble. Maintain at least 1 mm of reserve from both guide ends. The ideal 12-tooth polygon estimate is about 0.63 mm; use the recorded idler-travel range for the complete raised-end geometry; the physical chain also needs manufacturing and wear allowance. Set tension only high enough to retain engagement and check a full revolution by hand in both directions before powering it.

Recessed slew bearing and gear assembly

R2 lowers the upper deck by 48 mm while retaining the 40 mm bearing separation. The bearing tube and motor pass through the bridge; their mounting flanges still bear on the Z80 bridge top. Assemble the lower cap before fitting the housing to the chassis. The steel spindle and paired 6004 bearings carry the upper body; the motor drives the gears.

Item	Z interval, mm	Interface
Bottom bearing cap	28 to 32	Four M3 clearances on Ø54 at 45°
Lower outer-race spacer	32 to 44	OD42/ID35 × 12, steel
Lower 6004 bearing	44 to 56	20 × 42 × 12
Inner-race spacer	56 to 84	OD26/ID20.1 × 28, steel
Upper 6004 bearing	84 to 96	Bearing centres 40 mm apart
Upper outer-race spacer	96 to 102	OD42/ID35 × 6, steel
Upper bearing cap	102 to 106	Four M3 clearances on Ø54
Upper inner-race spacer	96 to 114	OD28/ID20.1 × 18, steel
Clamp flange	114 to 124	Ø20 clamp bore; OD70; four tapped M5 holes on Ø50 at 45°
Upper deck	124 to 134	Matching flange holes; flush screw heads

The Ø20 × 90 spindle spans Z34 to 124. The lower Ø20-bore collar spans Z34 to 44. Use accurately faced spacers and shims to remove end float without binding. Outer spacers contact outer races; the inner and top spacers contact inner races. Bearing envelopes do not model individual balls or establish preload. Four M3 × 85 tie bolts retain both caps. The Ø64 housing body has a Ø104 flange at Z80–88, secured through the bridge with four M5 bolts on Ø84 at 45°.

The module-1.5 gears have 96 and 20 teeth, 20° pressure angle and 10 mm face width. They occupy Z106–116. Their axes are 87.2 mm apart, including backlash allowance. Mount the ring on six 8 mm metal spacers using M4 × 30 countersunk screws and captured nuts. Mount the deck to the clamp flange with M5 × 20 countersunk screws. The pinion uses a purchased Ø4 shaft clamp hub, Ø24 × 7, with three M3 tapped holes on Ø18. Use M3 × 16 screws with 1 mm washers, and keep heads below Z124. Match or machine the metal hubs to the supplied STEP interfaces.

The vertical slew motor is centred at (87.2,0), with body Z46–97 and Ø4 shaft Z97–116. Its saddle has a top flange at Z80–86, with four M4 mounts at X58.2/116.2, Y±19. The motor passes through a clearance opening in the bridge. Formed steel straps clamp the body at Z54/84; fit them before enclosing the chassis. The four mounting bolts pass outside the motor opening into the bridge. Confirm shaft length and gearbox dimensions before purchase. Set backlash with a paper strip, remove it and check the complete gear cycle by hand.

The two Ø8 steel stop posts stand from Z80 to 124 at radius120 and ±15°. Their exposed length is 44 mm, plus the M5 mounting ends. The rear tab sits at Z117–124. Normal slew is at most ±150°; sampled hard-stop contact occurs around ±160.75°. Fit position feedback and independent normally-closed direction limits before the hard stops, with reverse motion allowed away from an active limit. Do not use impacts for homing.

The removable skirt rotates with the deck. Its lower edge is Z92: 12 mm above the bridge and clear of the tracks. Six M3 × 25 countersunk screws pass through the deck into captured nuts in its bosses. A notch clears the rear stop tab. Install the nuts before assembling the skirt. It encloses the mechanism and does not carry bearing loads. Its clearance does not establish flexible-cable clearance.

R3 upper body

The upper body follows the Cat 320 layout. The cab is at the front left, beside the boom foot. A tank housing and a stepped toolbox fill the front right. The engine hood spans the full width behind them, and a curved counterweight closes the rear. The deck is 308 mm wide. Its rear arc and the counterweight follow the 189 mm tail-swing radius (Cat 320: 2,830 mm at 1:15). A 12 mm perimeter rim (Z112–124) stiffens the deck and reads as the upper-frame side members. It is left off the boom-foot nose so the bucket can pass in low poses.

Every shell is screwed down. Standard screws pass through printed bolt tubes into hex nuts held in pockets under the deck (Z124–134). Fit each nut before its shell goes on; the screw pulls it up against the pocket roof. No screw tip may reach below Z124, where the slew stop posts pass underneath. `simulation/packaging.json` checks every screw's tip height and nut engagement.

Part	Fasteners	Notes
18 counterweight	4 × M4 × 80 countersunk	Heads flush at Z206; internal ribs clamp the ballast plate
19 engine hood	4 × M3 × 80 countersunk	Fit the exhaust stack first: one M3 × 8 from inside into the nut in its base
29 hood hatch	4 × M3 × 12 countersunk	Nuts captured in the four hood bosses; battery access
30 tank housing	2 × M3 × 80 countersunk	Open rear and bottom: room for wiring or a controller
31 toolbox	2 × M3 × 45 countersunk	Access step at Z152
23 equipment tray	4 × M3 × 45 countersunk	Through the four legs; carries the cradle, BMS and converters
33 cell cradle	2 × M3 × 8	Nuts in the tray underside pockets
25 battery retainer	2 × M3 × 20	Nuts slide into side slots in the cradle posts

Lay the 54 × 150 × 24 mm steel ballast plate (about 1.53 kg) on the deck at X=-167 to -113, Y±75, Z134–158. Put the 1 mm rubber strip on top. The counterweight's side ribs locate the plate; its central rib presses on the strip when the four M4 bolts are tightened. Do not substitute printed plastic: the stability calculation assumes steel.

Cab and arm

The cab is 120 × 70 × 118 mm including the roof, compared with R1's 110 × 40 × 108 mm. Its length, width and height follow estimates from the Cat 320 manufacturer drawing at the arm's 1:15 reference scale; see `references/CAB_PROPORTIONS.md` for the measurements and limitations. The source gives overall machine dimensions, not standalone factory cab dimensions. The remaining vehicle dimensions adapt to the full-size bearings and original arm interfaces.

R3 keeps the R2 cab design but moves it 23 mm forward and onto the machine's left, as on the Cat 320. It occupies X=-22 to 98, Y=84 to 154 and Z134 to 252. Four M3 threaded rods pass through roof, corner posts and deck at (-17,149), (82,149), (-17,89) and (82,89). Cut them to 130 mm; their lower ends engage the deck nuts and stop at Z125. Fit a washer and nut on the roof without crushing the print. The seat uses four M3 × 35 bolts. Each console uses one M3 × 45 socket screw in a 3 mm counterbore. All of these go into deck nuts. Print two consoles; the second sits 50 mm inboard. All console details are decorative.

The STEP assembly includes four 0.8 mm PET glazing panels. Cut their outlines from the separate glazing STEP files or SVG templates, attach to inside faces with thin adhesive strips, and confirm clearance to the tie posts. Fit glazing after the cab hardware. The roof remains removable. The cab is not a sealed enclosure.

Mount the unchanged arm60 base at (45,0,134) mm, using four M6 × 25 screws at X13/121, Y±65 and recessed underside nyloc nuts. Keep all original arm shafts, spacers and collars. Build four V6-U60 actuators from the sibling folder. The original arm and actuator files are unchanged.

Two-18650 battery bay

The cells sit on the equipment tray inside the engine hood, under the hatch. The R2 rear cassette is removed. The cradle (33) is sized from manufacturer datasheets rather than a generic envelope:

Datasheet	Diameter, mm	Length, mm
Samsung INR18650-25R (2013)	18.26–18.40	64.70–65.00
LG INR18650HG2 (2014)	18.0–18.5	64.8–65.2
Panasonic NCR18650B (2012)	18.5 maximum	65.3 maximum

The cradle accepts flat-top, unprotected cells Ø18.0–18.6 × 64.7–65.3 mm; the upper limits are 0.1 mm above the largest datasheet maximum. Use a matched pair rated at least 15 A continuous each, such as the 25R or HG2. The NCR18650B fits but is a low-current cell. Button-top or protected cells, 67–70 mm long, do not fit.

Have the pair spot-welded as a side-by-side 2S1P pack. It uses 0.15 mm pure-nickel strip (one series link at the +Y end, two terminal tabs at the -Y end), a 0.5 mm insulating separator between the cans, 0.3–1.0 mm end insulators and 0.12–0.25 mm PVC wrap. Never solder to the cans, and do not use the printed cradle as electrical contacts. The finished pack is at most 38.2 × 19.1 × 67.6 mm. Its leads leave through the notch in the -Y end wall.

Two Ø19.7 mm grooves on a 19.3 mm pitch locate the cells, with their axes along Y. The groove pair is centred at X=-68, with the groove axes at Z181.35. The end walls are 69 mm apart. Put two Ø17 × 4 mm closed-cell foam discs at the +Y end and the 3 mm foam pad over the pack. Then fit the retainer bar with two M3 × 20 screws into nuts slid into the side slots of the cradle posts. Across the datasheet range the end pads compress to 1.4–3.4 mm and the top pad to 1.3–2.1 mm, so the pack is always held but never crushed. The minimum radial clearance in the grooves is 0.2 mm. Try a real cell in the coupon's groove before printing the cradle.

Reserved component	X / Y envelope, mm	Height / installation
2S BMS	X -60 to -31; Y -78 to -48	Z173–179 on two insulating standoffs
6 V buck-boost converter	X -100 to -63; Y -78 to -46	Z173–189 on two insulating standoffs
12 V boost converter	X -100 to -40; Y 46 to 78	Z173–194 on two insulating standoffs

These are maximum module envelopes, not verified supplier parts. Each board has two M3 holes in the tray for insulating standoffs or an adapter plate matched to the actual board. Confirm hole patterns, underside clearance and cooling. The harness leaves the tray through the slot in its front edge (X-46 to -34). It reaches the chassis through the Ø8 deck opening at (-10,25) under the hood nose; fit a grommet there.

For service, switch off and unplug the pack. Remove the four M3 × 12 hatch screws and the two retainer screws, then lift the welded pack straight up. The BMS, converters and tray stay in place, and the cab and arm do not need removal. CAD checks lift the pack from 0 to 80 mm without collision, with about 14 mm clearance around it in the hatch opening.

Battery power and controls

Two 18650 cells in series provide about 7.2 V nominal and 8.4 V fully charged; they are not a direct 12 V supply. Use a 2S balancing/protection BMS rated at least 10 A continuous, matched cells rated at least 15 A continuous, a 10 A main fuse, master switch and accessible stop circuit. Remove the pack for charging with the matching 8.4 V CC/CV 2S charger. The model does not include a commissioned charger or control PCB.

The 12 V boost rail supplies travel and slew drivers; reserve up to 30 W. A regulated 6 V buck-boost rail supplies the four U60 actuators; reserve up to 18 W. Enforce a shared 36 W output budget and an 8 A pack current limit, so these rail maxima cannot be demanded simultaneously. At 6.2 V and assumed 85% converter efficiency, 36 W requires approximately 6.83 A. The controller cutoff is 6.2 V; BMS undervoltage protection is the backup. Rail limits, converter cooling, cell sag and motor current must be measured. The existing motor torque assumptions only hold if the regulated rails and current limits support the purchased motors. Begin with travel and arm motion separately. Do not infer torque or runtime from cell count alone.

The BMS and converters sit on the equipment tray beside the pack. Route the regulated 12 V feed and control/stop signals to lower-chassis travel/slew drivers through the deck opening at (-10,25). Provide a flexible service loop, strain relief and a grommet, with clearance from the gear, flange and spindle throughout the allowed swing. This is limited rotation with no slip ring. Test the cable loop with covers off before enclosing it.

Both boom cylinders require measured synchronisation. The inherited V6 end-stop boards do not provide position feedback. Use independent stroke sensing and stop both drives on excessive mismatch. Matching voltages alone does not establish equal stroke.

Boom and slew operation

Lowering the deck exposed a real self-collision: with the boom down, stick tucked in and upper body sideways, the bucket can hit a track. The unrestricted failing CAD case is retained in `simulation/unrestricted_lowered_body.json`; it has not been relabelled as a pass.

Boom stroke is available over the full 0–60 mm range at every slew angle within ±150°. The 15 mm / ±20° interlock has been removed: lowering the boom does not change slew, and slewing does not change boom extension. Track clearance is an operating responsibility.

The retained CAD clearance report samples a reference region: boom stroke at least 15 mm or slew within ±20°. Its passing results apply to those samples, not all combinations of the independent controls. The known bucket/track collision remains documented; the reference region is not enforced by the preview. No vehicle controller firmware is supplied.

Limit initial travel to 0.10 m/s, slew to 6°/s and motor torque/current to the verified drive requirements. The power-loss grade simulation rolls backward: powered holding is not a parking brake. Commission on a level restrained surface and use chocks for stationary tests.

Commissioning sequence

1. Print the fit coupon and two shoes. Check bearings, pins, steel sleeves, hinge articulation and retaining clips against the actual hardware.

2. Assemble each track without motors. Verify free wheels, guide travel, spring reserve and several hand-driven circuits in both directions.

3. Fit motors and couplings with tracks lifted. Check direction, low-speed drive, bushing entry/exit, current and wear in both directions.

4. Assemble the recessed slew bearing stack, empty deck and stop system. Verify backlash, feedback, independent limits and reversal away from a limit. Fit the skirt only after these checks.

5. Clamp the ballast under the counterweight, fit the tray, cradle and electronics, then the insulated two-cell pack. Measure regulated rail voltages, cutoff, current limiting and converter temperature with a current-limited supply before using the pack.

6. Install the arm and cab. Verify paired-cylinder synchronisation, feedback-loss inhibition and independent travel limits. Exercise boom and slew controls through clear paths, observing bucket/track clearance.

7. Weigh the subassemblies and update the mass model. Measure actuator thrust, motor torque/current, deflection, temperature, creep and endurance. Begin unloaded; the calculated 0.15 kg study load is not a tested payload rating.

8. Keep the bucket clear of a solid floor, or provide a digging recess. Progress to known loads only after measured margins support them. Record results before a general print release.

No physical print, fit, load, soil-digging, endurance or powered-hardware test has been performed. Digital results and assumptions are in `validation.json`, `simulation/` and `simulation_report.pdf`.